

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 30 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

*I Perarasu.S(192524024) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:Perarasu.S

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

Smart Home Automation System Using Intelligent Agents

A smart home automation system is designed to improve the comfort, safety, and energy efficiency of a house by automatically controlling devices such as lights, fans, air conditioners, heaters, and security systems. The system receives real-time information from sensors, including motion detectors, temperature sensors, smoke detectors, cameras, and clocks. It also learns the habits and preferences of users over time. Different types of intelligent agents can be used to achieve these goals.

1. Simple Reflex Agent

A Simple Reflex Agent makes decisions based only on the current percept without considering previous events. It follows predefined condition-action rules.

Working in the smart home:

- If motion is detected at night, the lights are switched on.
- If smoke is detected, the fire alarm is activated.
- If the door is opened unexpectedly, the security alarm sounds.

Advantages

- Fast response.
- Easy to design and implement.
- Suitable for simple repetitive tasks.

Disadvantages

- Cannot remember previous situations.
 - Cannot learn user preferences.
 - Performs poorly in complex environments.
-

2. Model-Based Reflex Agent

A Model-Based Reflex Agent maintains an internal model of the environment. It remembers previous states and predicts changes in the environment.

Working in the smart home:

- If nobody has been detected in a room for 20 minutes, the lights and fan are automatically switched off.
- If the room temperature has been increasing continuously, the air conditioner starts before it becomes uncomfortable.
- It knows whether doors were previously locked or unlocked.

Advantages

- Handles partially observable environments.
- Makes better decisions than simple reflex agents.
- Uses memory to improve performance.

Disadvantages

- More complex than simple reflex agents.
 - Requires additional memory and processing.
-

3. Goal-Based Agent

A Goal-Based Agent selects actions that help achieve a specific objective.

Working in the smart home:

Goals may include:

- Maintain room temperature at 24°C.
- Reduce electricity consumption.
- Keep all doors locked when the user leaves home.
- Ensure maximum home security.

The agent evaluates different actions and chooses the one that best satisfies these goals.

Advantages

- Flexible decision-making.
- Can plan multiple steps ahead.
- Suitable for dynamic environments.

Disadvantages

- Requires more computation.
 - Planning may take additional time.
-

4. Utility-Based Agent

A Utility-Based Agent chooses the action that provides the highest overall benefit (utility). It considers multiple objectives simultaneously.

Working in the smart home:

Suppose the user prefers a cool room but electricity prices are high.

The utility agent considers:

- User comfort
- Energy consumption
- Electricity cost
- Home security

Instead of running the AC continuously, it adjusts the temperature gradually to balance comfort and energy savings.

Similarly, during nighttime it may reduce unnecessary lighting while maintaining enough brightness for safety.

Advantages

- Produces the best overall decision.
- Balances multiple conflicting objectives.
- Suitable for real-world intelligent systems.

Disadvantages

- Complex to design.
 - Requires utility calculations and learning.
-

Comparison of Intelligent Agents

Agent Type	Memory	Goal-Oriented	Learns Preferences	Suitable for Smart Home
Simple Reflex	No	No	No	Basic Automation
Model-Based	Yes	No	Limited	Good
Goal-Based	Yes	Yes	Limited	Better
Utility-Based	Yes	Yes	Yes	Excellent

Recommended Approach

The most effective solution is a **Hybrid Intelligent Agent** combining:

- **Simple Reflex Agent** for immediate emergency responses.
- **Model-Based Agent** to remember environmental conditions.
- **Goal-Based Agent** to achieve comfort and security objectives.
- **Utility-Based Agent** to optimize energy consumption and user satisfaction.

Justification

The hybrid approach provides:

- Fast emergency response.
- Learning of user behavior.
- Intelligent planning.
- Maximum comfort.

- Better energy efficiency.
- Improved home security.

For example, the system learns that the user usually sleeps at 10:30 PM. It automatically dims the lights, adjusts the AC to the preferred temperature, locks all doors, and activates security cameras while minimizing electricity usage.

Conclusion

A Hybrid Intelligent Agent is the best choice because it combines quick reactions, memory, planning, learning, and optimization. It ensures comfort, safety, and efficient energy management, making it ideal for modern smart homes.

Question 2

Robot Navigation in an Unknown Maze Using Uninformed Search

A robot placed inside an unknown maze must find the exit without having prior knowledge of the maze layout. Since no information about the direction of the goal is available, uninformed search algorithms are suitable. Two important uninformed search strategies are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

Uniform Cost Search (UCS)

Uniform Cost Search always expands the node with the lowest cumulative path cost.

Working

1. Start from the initial position.
2. Store all possible paths in a priority queue.
3. Expand the least costly path.
4. Continue until the exit is found.

Example

Suppose the robot has two possible paths:

- Path A → Cost = 8
- Path B → Cost = 5

The UCS algorithm selects Path B first because it has the lower total cost.

Advantages

- Finds the least-cost path.
- Guarantees an optimal solution.
- Works well when movement costs differ.

Disadvantages

- High memory usage.
 - Can be slow for large mazes.
 - Stores many nodes in memory.
-

Iterative Deepening Search (IDS)

IDS combines the advantages of Depth-First Search and Breadth-First Search.

Working

The algorithm repeatedly performs depth-limited searches.

Example:

- Depth 0 → Start node only
 - Depth 1 → Explore one level
 - Depth 2 → Explore two levels
 - Continue until the exit is found.
-

Advantages

- Requires very little memory.
- Guarantees finding the solution if one exists.
- Suitable when the depth of the solution is unknown.

Disadvantages

- Repeats exploration of upper-level nodes.
- May take more time because of repeated searches.

Time and Space Complexity

Algorithm	Time Complexity	Space Complexity	Optimal
UCS	$O(b^{(1+C^*/\epsilon)})$ (depends on path costs)	High	Yes
IDS	$O(b^d)$	$O(bd)$	Yes (for equal step costs)

Where:

- **b** = branching factor
- **d** = depth of the goal
- **C*** = optimal path cost
- **ϵ** = minimum step cost

Relation to Intelligent Agents

Simple Reflex Agent

Acts only on current sensor input.

In the maze:

- Moves based only on nearby walls.
- Cannot remember previously visited locations.
- Often revisits the same paths.

Model-Based Agent

Maintains an internal map of explored locations.

In the maze:

- Remembers visited paths.
- Avoids unnecessary repetition.
- Makes more efficient decisions.

Goal-Based Agent

Its goal is to reach the exit.

The robot evaluates different paths and chooses actions that move closer to achieving the goal.

Utility-Based Agent

Considers several factors:

- Distance
- Time
- Energy consumption
- Safety

It selects the path with the highest overall benefit rather than simply the shortest path.

Effect of Agent Design on Decision-Making

The quality of decisions depends on the type of agent used.

- Simple Reflex Agents make quick but limited decisions.
- Model-Based Agents improve navigation using memory.
- Goal-Based Agents plan toward the exit.
- Utility-Based Agents optimize efficiency by balancing cost, time, and energy.

As the intelligence of the agent increases, navigation becomes more efficient and reliable.

Best Combination

The most effective combination is a **Model-Based Goal-Based Agent using Uniform Cost Search**.

Justification

- The **Model-Based Agent** remembers explored paths.
- The **Goal-Based Agent** focuses on reaching the exit.
- **Uniform Cost Search** guarantees the least-cost path when movement costs vary.

If all movement costs are equal and memory is limited, a **Goal-Based Agent with Iterative Deepening Search** is also an excellent choice because IDS uses much less memory.

Conclusion

For an unknown maze, both UCS and IDS are effective uninformed search strategies. UCS is preferred when finding the minimum-cost path is important, while IDS is better when memory is limited. Combining a Model-Based and Goal-Based Agent with UCS provides the most intelligent, efficient, and reliable solution for robot navigation because it uses memory, planning, and optimal path selection to reach the exit successfully.